

Gravitation

One law for falling apples and orbiting moons: Newton's gravitation, orbits, escape speed and Kepler's laws.

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Until now, “gravity” meant a constant downward force mg . That is fine near the Earth’s surface, but satellites, planets, tides and space missions need the full law. Newton’s insight was that the force making an apple fall is the same force that holds the Moon in orbit, weakening with distance as $1/r^2$. This chapter develops the law of universal gravitation, gravitational potential energy and escape speed, the physics of orbits, and Kepler’s laws. These are the working tools of satellite engineering, GPS, space medicine and astrophysics.

YOU WILL BE ABLE TO

- Apply Newton’s law of gravitation, with superposition, to systems of masses.
- Use the shell theorem, and calculate how g varies with altitude, depth and latitude.
- Derive $U = -GMm/r$, and use it to find escape speeds and energies.
- Analyse circular orbits: speed, period, energy, and geostationary orbits.
- State and use Kepler’s three laws, including conservation of angular momentum in elliptical orbits.
- Estimate tidal effects and understand apparent weightlessness.

THINK FIRST

The International Space Station orbits just 400 km above the Earth, where gravity is still about 89% as strong as on the ground. So why do astronauts float?

Because they are **falling**, and so is the station around them. Both are in free fall towards the Earth, with the same acceleration. They also move sideways at about 7.7 km/s, so they keep falling *around* the Earth instead of into it. Nothing pushes on the astronauts relative to the station, so their apparent weight is zero (Chapter 3). Newton imagined a cannon on a mountain firing balls faster and faster: at a high enough speed, the ball would fall all the way round the Earth without ever landing. That is an orbit.

§8.1

Newton's law of universal gravitation

Every pair of point masses attracts each other with a force along the line joining them:

$$F = \frac{Gm_1m_2}{r^2}, \quad G = 6.674 \times 10^{-11} \text{ N m}^2/\text{kg}^2.$$

- The force is always **attractive**, and it obeys Newton's third law: m_1 pulls m_2 exactly as hard as m_2 pulls m_1 .
- Gravitational forces obey **superposition**: the force on a mass is the vector sum of the forces from all the other masses.
- G is tiny. Gravity between everyday objects is minute, and it dominates only when at least one mass is astronomically large. Henry Cavendish first measured G in 1798 with a torsion balance, an experiment often called “weighing the Earth”.

KEY IDEA

Shell theorem (proved by Newton with calculus).

1. A uniform spherical shell attracts a mass *outside* it as if all the shell's mass were concentrated at its centre.
2. A uniform shell exerts **no net force** on a mass *inside* it.

So a spherically symmetric planet or star acts, from outside, like a point mass at its centre.

WORKED EXAMPLE 8.1**Gravity between people, and between Earth and Moon**

(a) Find the gravitational force between two 70 kg people standing 1.0 m apart. (b) Find the force between the Earth (5.97×10^{24} kg) and the Moon (7.35×10^{22} kg), whose centres are 3.84×10^8 m apart.

Solution

(a) For the two people:

$$F = \frac{(6.674 \times 10^{-11})(70)(70)}{1.0^2} = 3.3 \times 10^{-7} \text{ N.}$$

That is about the weight of a grain of fine sand. It is completely overwhelmed by friction.

(b) For the Earth and Moon:

$$F = \frac{(6.674 \times 10^{-11})(5.97 \times 10^{24})(7.35 \times 10^{22})}{(3.84 \times 10^8)^2} = \frac{2.93 \times 10^{37}}{1.47 \times 10^{17}} = 2.0 \times 10^{20} \text{ N.}$$

§8.2

The acceleration due to gravity

At the surface of a spherical planet of mass M and radius R , the weight of a mass m is $mg = GMm/R^2$, so

$$g = \frac{GM}{R^2}.$$

Measuring g , R and G therefore gives the mass of the Earth:

$$M = gR^2/G = 9.8(6.371 \times 10^6)^2/6.674 \times 10^{-11} = 5.96 \times 10^{24} \text{ kg.}$$

With altitude. At height h above the surface:

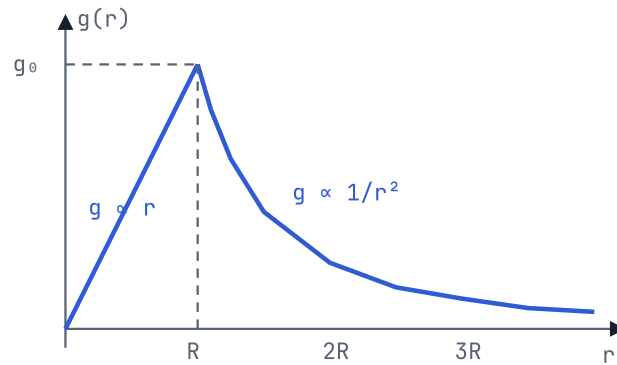
$$g(h) = \frac{GM}{(R+h)^2} = g \left(\frac{R}{R+h} \right)^2 \approx g \left(1 - \frac{2h}{R} \right) \quad \text{for } h \ll R.$$

With depth. Inside a uniform sphere, at radius $r < R$, only the mass inside radius r pulls (by the shell theorem). That mass is $M(r/R)^3$, so

$$g(r) = \frac{GM(r/R)^3}{r^2} = g \frac{r}{R}.$$

Inside, g falls *linearly* to zero at the centre. The real Earth has a dense core, so g actually increases slightly as you go down through the mantle.

With latitude. The Earth's rotation reduces the *effective* g at the equator by $\omega^2 R = 0.034 \text{ m/s}^2$, because part of the gravitational pull supplies the centripetal force. The Earth's equatorial bulge adds a further reduction. Altogether, g ranges from about 9.780 m/s^2 at the equator to 9.832 m/s^2 at the poles.



For a uniform planet, g grows linearly with r inside and falls as $1/r^2$ outside, peaking at the surface.

IN PRACTICE

Gravimetry. Tiny variations in g , of a few parts in 10^8 , reveal what lies underground. Dense ore bodies raise g slightly, and salt domes, oil-bearing rock and buried cavities lower it. Geophysicists map these anomalies with portable gravimeters to explore for minerals and oil, to detect sinkholes before construction, and even to monitor volcanoes as magma moves. From orbit, the GRACE satellites measured changes in the Earth's gravity field caused by groundwater depletion and melting ice sheets. Mass that moves, moves g .

§8.3

Gravitational potential energy

The formula $U = mgy$ assumes that g is constant. For large distances we need the exact form. The work done by gravity on a mass m moving radially from r_1 to r_2 is

$$W = \int_{r_1}^{r_2} \left(-\frac{GMm}{r^2} \right) dr = GMm \left(\frac{1}{r_2} - \frac{1}{r_1} \right).$$

Since $W = -\Delta U$, and choosing $U = 0$ at $r = \infty$:

$$U(r) = -\frac{GMm}{r}.$$

The potential energy is **negative** everywhere, and it rises towards zero as $r \rightarrow \infty$. A bound system has negative total energy: you must *add* energy to separate the masses. For small heights near the surface, $\Delta U = GMm \left(\frac{1}{R} - \frac{1}{R+h} \right) \approx \frac{GMm}{R^2} h = mgh$. ✓

The **gravitational potential** is the potential energy per unit mass, $V = U/m = -GM/r$, measured in J/kg.

Escape speed

The **escape speed** is the minimum launch speed for a projectile to reach $r = \infty$ (with zero speed there), ignoring air resistance. By energy conservation:

$$\frac{1}{2}mv_{\text{esc}}^2 - \frac{GMm}{R} = 0 \Rightarrow v_{\text{esc}} = \sqrt{\frac{2GM}{R}} = \sqrt{2gR}.$$

It does not depend on the mass of the projectile, or on the direction of launch (provided the path does not hit the planet).

WORKED EXAMPLE 8.2

Escaping the Earth and the Moon

Find the escape speed from (a) the Earth and (b) the Moon (mass 7.35×10^{22} kg, radius 1.737×10^6 m).

Solution

(a) Earth:

$$v_{\text{esc}} = \sqrt{2gR} = \sqrt{2(9.8)(6.371 \times 10^6)} = 1.12 \times 10^4 \text{ m/s} = 11.2 \text{ km/s}.$$

(b) Moon:

$$v_{\text{esc}} = \sqrt{\frac{2(6.674 \times 10^{-11})(7.35 \times 10^{22})}{1.737 \times 10^6}} = \sqrt{5.65 \times 10^6} = 2.38 \text{ km/s}.$$

Evaluate. The Moon's low escape speed is why it has no atmosphere. At the sunlit surface temperature, a small fraction of gas molecules move faster than 2.4 km/s, and over

billions of years they leak away. It is also why the Apollo lunar modules could return to orbit with a small engine.

§8.4

Circular orbits

For a satellite of mass m in a circular orbit of radius r around a much larger mass M , gravity provides the centripetal force:

$$\frac{GMm}{r^2} = \frac{mv^2}{r}.$$

This gives the orbital speed and period:

$$v = \sqrt{\frac{GM}{r}}, \quad T = \frac{2\pi r}{v} = 2\pi \sqrt{\frac{r^3}{GM}}.$$

The second result is **Kepler's third law** for circular orbits: $T^2 \propto r^3$. Both v and T are independent of the satellite's mass.

Energy in a circular orbit.

$$K = \frac{1}{2}mv^2 = \frac{GMm}{2r}, \quad U = -\frac{GMm}{r}, \quad E = K + U = -\frac{GMm}{2r}.$$

So $K = -E$ and $U = 2E$. A satellite that loses energy, for example to atmospheric drag, drops to a *lower* orbit and **speeds up**, because $K = -E$ increases as E becomes more negative.

WORKED EXAMPLE 8.3

The International Space Station

The ISS orbits 400 km above the Earth's surface. Find its speed and period. Use $GM_E = 3.986 \times 10^{14} \text{ m}^3/\text{s}^2$.

Solution

$$r = 6.371 \times 10^6 + 4.0 \times 10^5 = 6.771 \times 10^6 \text{ m}.$$

$$v = \sqrt{\frac{3.986 \times 10^{14}}{6.771 \times 10^6}} = \sqrt{5.89 \times 10^7} = 7.67 \times 10^3 \text{ m/s (27 600 km/h)}.$$

$$T = \frac{2\pi r}{v} = \frac{2\pi(6.771 \times 10^6)}{7670} = 5.55 \times 10^3 \text{ s} = 92 \text{ min.}$$

The crew see about 16 sunrises a day.

WORKED EXAMPLE 8.4

The geostationary orbit

A geostationary satellite stays above the same point on the equator, so it must orbit once per *sidereal* day (86 164 s, the Earth's rotation period relative to the stars). Find its orbital radius and altitude.

Solution

From Kepler's third law:

$$r^3 = \frac{GMT^2}{4\pi^2} = \frac{(3.986 \times 10^{14})(86\,164)^2}{4\pi^2} = \frac{2.959 \times 10^{24}}{39.48} = 7.50 \times 10^{22} \text{ m}^3.$$

$$r = 4.22 \times 10^7 \text{ m}, \quad \text{altitude} = 4.22 \times 10^7 - 6.37 \times 10^6 = 3.58 \times 10^7 \text{ m} \approx 35\,800 \text{ km.}$$

Evaluate. There is only one such orbit: one radius, in the equatorial plane. It is a crowded and valuable resource, used by communications and weather satellites, whose dishes on the ground can stay fixed. Light takes about 0.12 s to reach it and the same to come back, which causes the delay you notice on satellite telephone calls.

WORKED EXAMPLE 8.5

The energy cost of raising an orbit

How much energy is needed to move a 1000 kg satellite from the ISS orbit ($r_1 = 6.771 \times 10^6 \text{ m}$) to a geostationary orbit ($r_2 = 4.22 \times 10^7 \text{ m}$)?

Solution

$$\Delta E = -\frac{GMm}{2r_2} + \frac{GMm}{2r_1} = \frac{GMm}{2} \left(\frac{1}{r_1} - \frac{1}{r_2} \right).$$

$$\Delta E = \frac{(3.986 \times 10^{14})(1000)}{2} (1.477 \times 10^{-7} - 2.37 \times 10^{-8}) = 1.993 \times 10^{17} \times 1.240 \times 10^{-7}$$

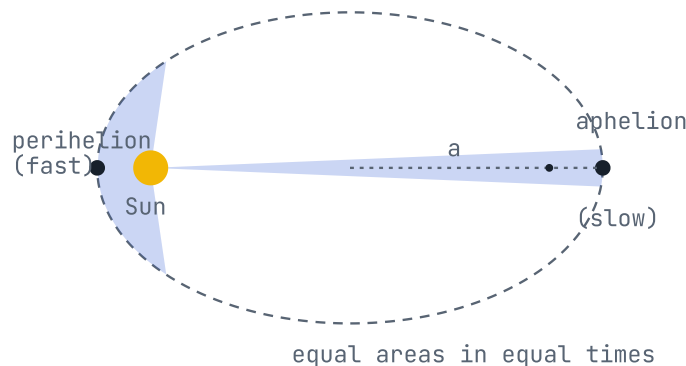
Evaluate. The satellite *slows down*, from 7.7 km/s to 3.1 km/s, yet its total energy increases, because its potential energy rises by twice as much as its kinetic energy falls. Rockets deliver this energy in two engine burns, as Problem P8.14 explores.

§8.5

Kepler's laws and elliptical orbits

Johannes Kepler (1609–1619) extracted three empirical laws from Tycho Brahe's planetary observations. Newton showed that all three follow from the $1/r^2$ law.

1. **Law of orbits.** Each planet moves in an ellipse with the Sun at one focus. More generally, bound orbits are ellipses, and circles are a special case. Unbound orbits are parabolas or hyperbolas.
2. **Law of areas.** The line from the Sun to a planet sweeps out equal areas in equal times.
3. **Law of periods.** $T^2 = \frac{4\pi^2}{GM} a^3$, where a is the semi-major axis of the ellipse.



Kepler's second law. The two shaded sectors have equal areas and are swept in equal times, so the planet moves fastest at perihelion.

The law of areas is conservation of angular momentum. Gravity always points towards the Sun, so it exerts no torque about the Sun, and $\mathbf{L} = m\mathbf{r}\mathbf{v}_\perp$ is constant. In a short time $d\mathbf{t}$ the line to the planet sweeps a thin triangle of area $dA = \frac{1}{2}r(v_\perp dt)$, so

$$\frac{dA}{dt} = \frac{1}{2}rv_\perp = \frac{L}{2m} = \text{constant}.$$

At perihelion and aphelion the velocity is perpendicular to the radius, so $r_p v_p = r_a v_a$.

Energy of an elliptical orbit. The result for circles generalises: $E = -\frac{GMm}{2a}$, where a is the semi-major axis. Combined with $E = \frac{1}{2}mv^2 - GMm/r$, this gives the vis-viva equation:

$$v^2 = GM \left(\frac{2}{r} - \frac{1}{a} \right).$$

Mission planners use it constantly.

WORKED EXAMPLE 8.6

Weighing Jupiter

Jupiter's moon Io orbits at 4.22×10^8 m from Jupiter's centre, with a period of 1.769 days. Find Jupiter's mass.

Solution

$T = 1.769 \times 86\,400 = 1.528 \times 10^5$ s. From Kepler's third law:

$$M = \frac{4\pi^2 a^3}{GT^2} = \frac{39.48 (4.22 \times 10^8)^3}{(6.674 \times 10^{-11})(1.528 \times 10^5)^2} = \frac{39.48 \times 7.52 \times 10^{25}}{1.558} = 1.90 \times 10^{27} \text{ kg.}$$

That is about 318 Earth masses.

Evaluate. This is how we know the masses of planets, stars and galaxies: watch something orbit, measure its period and orbital size, and apply Kepler's third law. Applied to stars orbiting the centre of our galaxy, the same method reveals a black hole of four million solar masses.

WORKED EXAMPLE 8.7

Halley's comet

Halley's comet has its perihelion at 0.586 AU from the Sun and its aphelion at 35.1 AU, where $1 \text{ AU} = 1.496 \times 10^{11}$ m. At perihelion it moves at 54.5 km/s. Find (a) its speed at aphelion and (b) its orbital period.

Solution

(a) **Angular momentum** is conserved, and at both ends the velocity is perpendicular to the radius:

$$v_a = v_p \frac{r_p}{r_a} = 54.5 \times \frac{0.586}{35.1} = 0.91 \text{ km/s.}$$

(b) **Kepler's third law.** For orbits around the Sun, $T^2 = a^3$ when T is in years and a in AU (calibrated by the Earth: 1 year, 1 AU). The semi-major axis is $a = (0.586 + 35.1)/2 = 17.8$ AU, so

$$T = a^{3/2} = 17.8^{1.5} = 75 \text{ years.}$$

Evaluate. The comet moves 60 times faster at perihelion than at aphelion, so it spends most of its 75 years far from the Sun, out beyond Neptune.

IN PRACTICE

Space engineering and space medicine.

- **GPS satellites** orbit at about 20 200 km with a period of half a sidereal day (Problem P8.11). Each carries atomic clocks. The receiver in your phone finds its position from the signal travel times from at least four satellites. The system would drift by about 10 km per day if it did not correct for both special and general relativity.
- **Microgravity medicine.** In orbit, the body is no longer loaded by its weight. Astronauts lose 1–2% of the bone mass in weight-bearing bones per month, their muscles atrophy, and body fluids shift towards the head, causing facial puffiness and changes in the eyes and optic nerve. Daily resistance exercise on special machines counters some of the loss. Rotating space stations that create “artificial gravity” through centripetal acceleration (Problem P8.12) are a long-standing proposal for missions to Mars.

§8.6

Tides

The Moon's gravity is slightly stronger on the side of the Earth facing it, and slightly weaker on the far side, than at the Earth's centre. Relative to the centre, the near side is pulled towards the Moon and the far side is “left behind”. This produces two tidal bulges, and so two high tides a day. The difference in gravitational acceleration across a body of size Δr at distance d from a mass M is

$$\Delta a \approx \frac{2GM}{d^3} \Delta r.$$

Because of the $1/d^3$ dependence, the nearby Moon raises tides about 2.2 times larger than the Sun does, even though the Sun pulls on the Earth 180 times harder (Problem P8.17). When the Sun and

Moon line up, at new and full moon, their tides add to give large *spring tides*. At quarter moons they partly cancel, giving small *neap tides*.

CHAPTER SUMMARY

- $F = Gm_1m_2/r^2$, with $G = 6.674 \times 10^{-11} \text{ N m}^2/\text{kg}^2$. Superposition applies. A spherical body acts as a point mass from outside, and a shell exerts no force inside.
- $g = GM/R^2$. $g(h) = g(R/(R+h))^2$ outside, and $g(r) = gr/R$ inside a uniform sphere.
- $U = -GMm/r$, with zero at infinity. Escape speed $v_{\text{esc}} = \sqrt{2GM/R}$.
- Circular orbits: $v = \sqrt{GM/r}$, $T = 2\pi\sqrt{r^3/GM}$ and $E = -GMm/2r$. Geostationary radius $4.22 \times 10^7 \text{ m}$.
- Kepler: ellipses with the Sun at a focus; equal areas in equal times (conservation of L , so $r_p v_p = r_a v_a$); $T^2 = 4\pi^2 a^3/GM$. For ellipses, $E = -GMm/2a$.
- Tidal acceleration $\approx 2GM\Delta r/d^3$.

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P8.1

The Earth pulls on the Moon with a force of $2 \times 10^{20} \text{ N}$. Why doesn't the Moon fall into the Earth? Or is it, in some sense, falling?

P8.2

Astronauts on the ISS are often described as being in “zero gravity”. Explain why this is wrong, and give a better description.

P8.3

Give two separate physical reasons why the measured value of g is smaller at the equator than at the poles.

P8.4

A satellite in a low circular orbit slowly loses energy to atmospheric drag. Does it speed up or slow down? Explain this apparent paradox using the energy relations for circular orbits.

LEVEL B – STANDARD PROBLEMS**P8.5**

Mars has a mass of 6.42×10^{23} kg and a radius of 3.39×10^6 m. Find g at its surface, and the weight there of an 80 kg astronaut in a 40 kg spacesuit.

P8.6

At what altitude above the Earth's surface is g half its surface value? Express your answer in km and as a fraction of the Earth's radius.

P8.7

Three 2.0 kg spheres sit at the corners of an equilateral triangle with 1.0 m sides. Find the magnitude and direction of the net gravitational force on one of them.

P8.8

At what point on the line between the Earth and the Moon is the net gravitational force on a spacecraft zero? (The Earth's mass is 81.3 times the Moon's.)

P8.9

A satellite orbits 2000 km above the Earth's surface. Find its speed and period.

P8.10

A neutron star has a mass of 2.8×10^{30} kg (1.4 solar masses) and a radius of 10 km. Find (a) the gravitational acceleration at its surface and (b) the escape speed, as a fraction of the speed of light.

P8.11

A GPS satellite orbits the Earth twice per sidereal day (period 43 082 s). Find its orbital radius, altitude and speed.

P8.12

Artificial gravity. A space station is designed as a large ring of radius 100 m that rotates to give 1.0 g of apparent gravity at the rim. (a) Find the rotation rate, in rad/s and rpm. (b) For an astronaut 2.0 m tall standing on the rim, find the percentage difference in apparent gravity between feet and head. (c) Explain why physiologists worry about rotation rates above about 2–4 rpm, and how a larger radius helps.

LEVEL C – CHALLENGE PROBLEMS**P8.13**

Gravity train. Imagine a straight tunnel drilled through the centre of a uniform Earth. A capsule is dropped in from rest at the surface. Ignore friction and the Earth's rotation. (a) Show that the force on the capsule is $\mathbf{F} = -(mg/R)\mathbf{r}$, where $\mathbf{r}$ is its distance from the centre. (b) Show that it oscillates with period $2\pi\sqrt{R/g}$, and evaluate this in minutes. (c) Find its maximum speed. (d) Compare with the period of a satellite skimming the surface, and explain the coincidence.

P8.14

Hohmann transfer. A satellite in a circular orbit of radius $\mathbf{r}_1 = 6.671 \times 10^6$ m (300 km altitude) is to be moved to the geostationary radius $\mathbf{r}_2 = 4.22 \times 10^7$ m. It uses an elliptical transfer orbit with perigee $\mathbf{r}_1$ and apogee $\mathbf{r}_2$, and two short engine burns. Using the vis-viva equation, find (a) the speed change needed at perigee, (b) the speed change needed at apogee, (c) the total $\Delta\mathbf{v}$, and (d) the time spent on the transfer orbit.

P8.15

An asteroid is released from rest at a distance of $2\mathbf{R}_E$ from the Earth's centre, one Earth radius above the surface. Ignore the atmosphere. (a) Find its speed when it hits the surface. (b) Compare this with the (wrong) result from $\mathbf{v} = \sqrt{2gh}$, and explain the difference. (c) What is the impact speed for an object falling from rest at a very large distance?

P8.16

Binary stars. Two stars, each of mass $\mathbf{M}$, orbit their common centre of mass in a circle, a distance $\mathbf{d}$ apart. (a) Show that the period is $\mathbf{T} = 2\pi\sqrt{\mathbf{d}^3/(2\mathbf{GM})}$. (b) Evaluate $\mathbf{T}$ for two solar-mass stars ($\mathbf{M} = 1.99 \times 10^{30}$ kg) separated by 1 AU, and compare it with one year.

P8.17

Tides. (a) Show that the difference between the gravitational accelerations produced by a mass M at distance d , at two points separated by Δr along the line to M , is approximately $2GM\Delta r/d^3$. (b) Use this with $\Delta r = R_E$ to compare the tidal accelerations due to the Moon (7.35×10^{22} kg at 3.84×10^8 m) and the Sun (1.99×10^{30} kg at 1.496×10^{11} m). (c) Explain spring and neap tides.

P8.18

Black holes. (a) Setting the escape speed equal to c , show that a mass M becomes a black hole if it is compressed within the radius $R_s = 2GM/c^2$. (This Newtonian argument happens to give the correct result from general relativity.) Evaluate R_s for the Sun and for the Earth. (b) Using the tidal formula, estimate the difference in gravitational acceleration between the head and the feet of a 2.0 m astronaut at R_s for (i) a 10-solar-mass black hole and (ii) the 4×10^6 -solar-mass black hole at the centre of our galaxy. Which could an astronaut survive approaching?

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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